

Final Exam Questions Spring 2025

1. ADC: Ihzaz Abrar Sadik
2. DAC: Ihzaz Abrar Sadik
3. Schmitt Trigger: Md. Zayed Bin Zahir Arju
4. Signal Generator: Nur Mohammad Shaon
5. NMOS+CMOS: Tasneem Mazhar

Please submit your questions by Sunday, 11 May 2025. Please make sure you make Set A and Set B of your question. Please provide a solution with the marks distribution to your question.

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- While posting your question here, please follow this format:

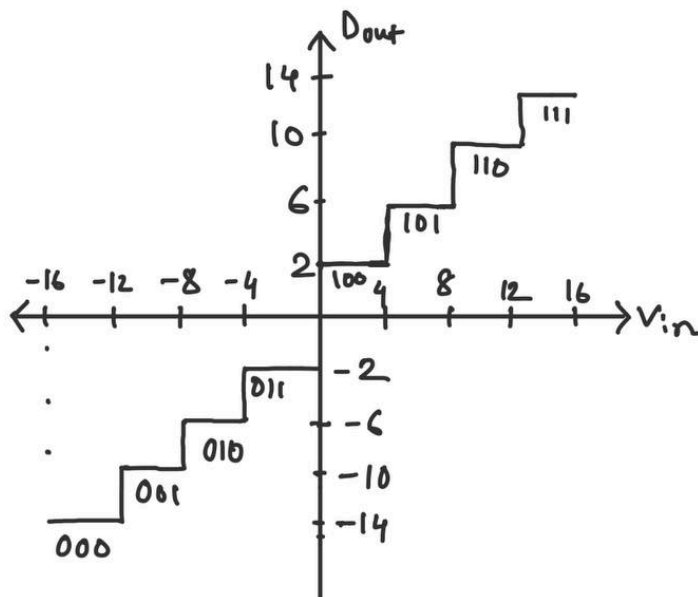
(a)	CO	Question	[Marks]
(b)	CO	Question	[Marks]
(c)	CO	Question	[Marks]

Question : ADC+DAC [ZAZ]

Set-A

$$\begin{aligned} a) \quad V_3 &= V_0 + 3 \times \Delta \\ &= -V_{ref} + 3 \frac{V_{ref} - (-V_{ref})}{8} \\ \Rightarrow -4 &= -V_{ref} + 3 \frac{V_{ref}}{4} \\ \Rightarrow V_{ref} &= 16V \end{aligned}$$

D_{out} vs V_{in} :

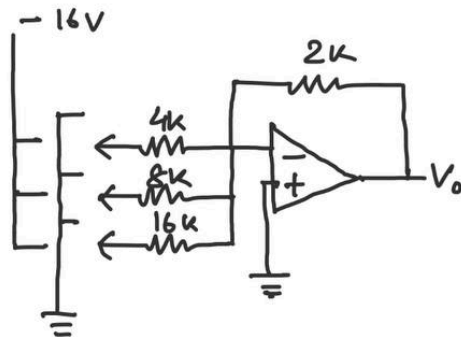


b) Here, $V_{ref} \times R_f \left(\frac{1}{R_1} + \frac{1}{2R_1} + \frac{1}{4R_1} \right) = 14$

$$\Rightarrow 16 \times 2 \times \frac{1}{R_1} \times 1.25 = 14$$

$$\Rightarrow R_1 = 4 \text{ k}\Omega$$

$$\therefore R_2 = 2R_1 = 8 \text{ k}\Omega, R_3 = 4R_1 = 16 \text{ k}\Omega$$



c) $V_{in} = \frac{m}{2^n} \times V_{ref}$
 $m = \text{decimal of } (10110101)_2$
 $= 181$

$$= \frac{181}{2^8} \times 16$$

$$= 11.3125 \text{ V}$$

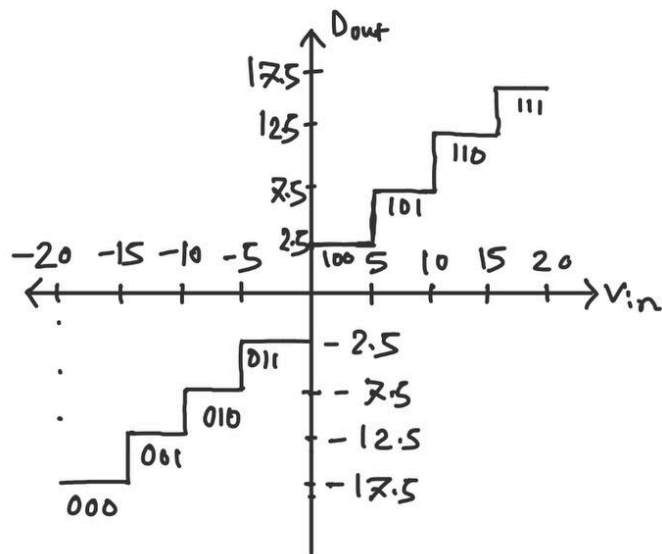
$$t_1 = \frac{2^n}{f_{clk}} = \frac{2^8}{1.5 \times 10^6} = 0.171 \text{ ms}$$

$$t_2 = \frac{2^n}{f_{clk}} \times \frac{m}{2^n} = 0.121 \text{ ms}$$

Set-B

$$\begin{aligned}
 a) \quad V_3 &= V_0 + 5 \times \Delta \\
 &= -V_{ref} + 5 \frac{V_{ref} - (-V_{ref})}{8} \\
 \Rightarrow 5 &= -V_{ref} + 5 \frac{V_{ref}}{4} \\
 \Rightarrow V_{ref} &= 20
 \end{aligned}$$

D_{out} vs V_{in} :

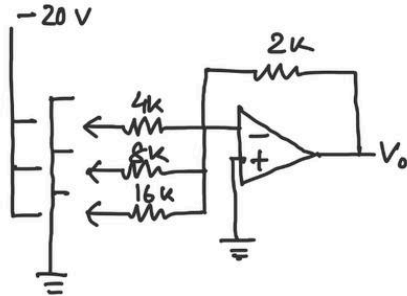


b) Here, $V_{ref} \times R_f \left(\frac{1}{R_1} + \frac{1}{2R_1} + \frac{1}{4R_1} \right) = 17.5$

$$\Rightarrow 20 \times 2 \times \frac{1}{R_1} \times 1.25 = 17.5$$

$$\Rightarrow R_1 = 4 \text{ k}\Omega$$

$$\therefore R_2 = 2R_1 = 8 \text{ k}\Omega, R_3 = 4R_1 = 16 \text{ k}\Omega$$



c) $V_{in} = \frac{m}{2^n} \times V_{ref} \quad \left| \quad \begin{array}{l} m = \text{decimal of} \\ (11010110)_2 \\ = 214 \end{array} \right.$

$$= \frac{214}{2^8} \times 20 = 0.832 \text{ V}$$

$$= 16.71875 \text{ V}$$

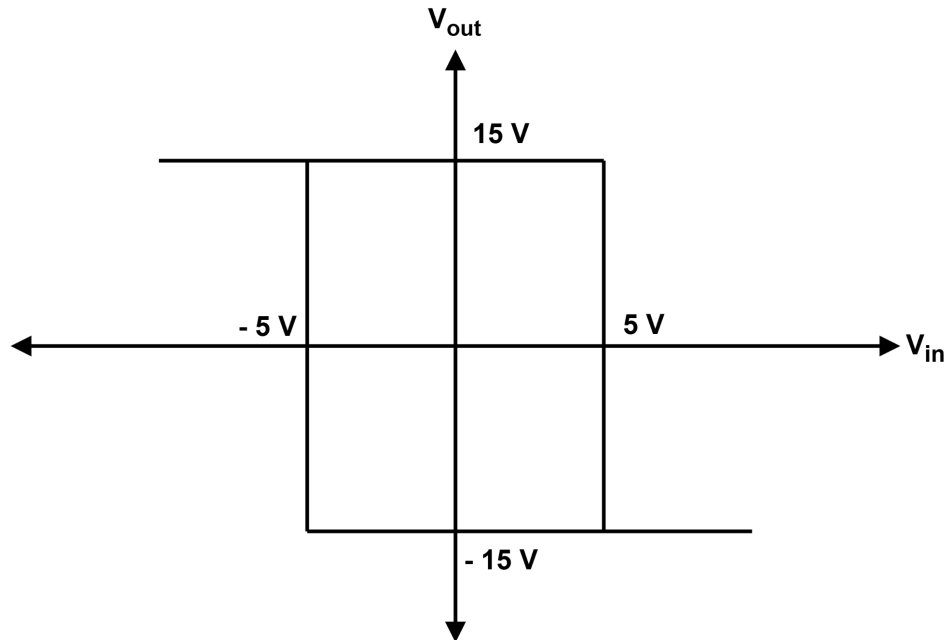
$$t_1 = \frac{2^n}{f_{clk}} = \frac{2^8}{2.5 \times 10^6} = 0.1024 \text{ ms}$$

$$t_2 = \frac{2^n}{f_{clk}} \times \frac{m}{2^n} = 0.0856 \text{ ms}$$

$$\therefore T_{conv} = t_1 + t_2 = 0.188 \text{ ms}$$

Question : Schmitt Trigger [ZBH]

Set-A



(a)	CO1	Identify the Schmitt Trigger Circuit from the given VTC. What is the hysteresis width V_{HW} , shift voltage V_S , V_H , V_L , V_{TH} and V_{TL} for this specific circuit.	1+2
(b)	CO1	Design a Schmitt Trigger Circuit that will produce the same VTC.	5
(c)	CO1	Now suppose the shift voltage, $V_S = +10\text{ V}$; draw the modified VTC. Now from the modified VTC, again design the Schmitt Trigger Circuit for this case.	6
(d)	CO1	Apply an input signal $V_{in}(t) = -12\sin(2\pi ft)$ into the Schmitt Trigger and VTC derived at (c) and draw/plot the output V_{out}	6

(a)

The following VTC belongs to inverting schmitt trigger with reference grounded,

$$\text{Hysteresis width, } V_{HW} = V_{TH} - V_{TL} = 5 - (-5) = 10 \text{ V}$$

$$\text{shift voltage, } V_S = \frac{V_{TH} + V_{TL}}{2} = \frac{5 + (-5)}{2} = 0 \text{ V}$$

$$V_H = 15 \text{ V} ; V_L = -15 \text{ V} ; V_{TH} = 5 \text{ V} ; V_{TL} = -5 \text{ V}$$

(b)

we know, for inverting schmitt trigger with grounded reference ($V_{ref} = 0 \text{ V}$) $\rightarrow$

$$V_{TH} = V_H \times \frac{R_1}{R_1 + R_2} \dots \textcircled{i} \quad \text{and} \quad V_{TL} = V_L \times \frac{R_1}{R_1 + R_2} \dots \textcircled{ii}$$

$$\textcircled{i} - \textcircled{ii} \Rightarrow (V_{TH} - V_{TL}) = (V_H - V_L) \frac{R_1}{R_1 + R_2}$$
$$\Rightarrow \{5 - (-5)\} = \{15 - (-15)\} \frac{R_1}{R_1 + R_2}$$

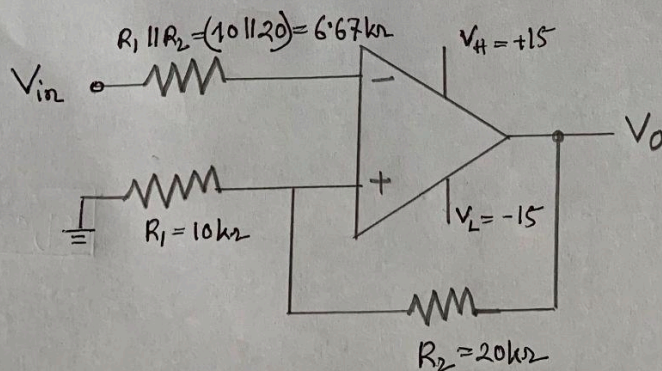
$$\Rightarrow \frac{10}{30} = \frac{R_1}{R_1 + R_2} = \frac{1}{3}$$

$$\Rightarrow R_1 + R_2 = 3R_1$$

$$\Rightarrow \boxed{R_2 = 2R_1}$$

assume, $R_1 = 10 \text{ k}\Omega$ then $R_2 = 20 \text{ k}\Omega$
(suppose)

design



(c) with $V_S = +10V$, the VTC will have right shift,

now, as there's shift voltage, so this 'inverting schmitt trigger' has ' V_{Ref} '

we know,

$$V_{TH} = V_{Ref} \times \frac{R_2}{R_1 + R_2} + V_H \times \frac{R_1}{R_1 + R_2} \dots \textcircled{i}$$

$$V_{TL} = V_{Ref} \times \frac{R_2}{R_1 + R_2} + V_L \times \frac{R_1}{R_1 + R_2} \dots \textcircled{ii}$$

$$\textcircled{i} - \textcircled{ii} \Rightarrow (V_{TH} - V_{TL}) = (V_H - V_L) \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow (15 - 5) = (15 - (-15)) \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow \frac{10}{30} = \frac{R_1}{R_1 + R_2} = \frac{1}{3}$$

$$\Rightarrow \boxed{2R_1 = R_2}$$

suppose, $R_1 = 10k\Omega$, then $R_2 = 20k\Omega$

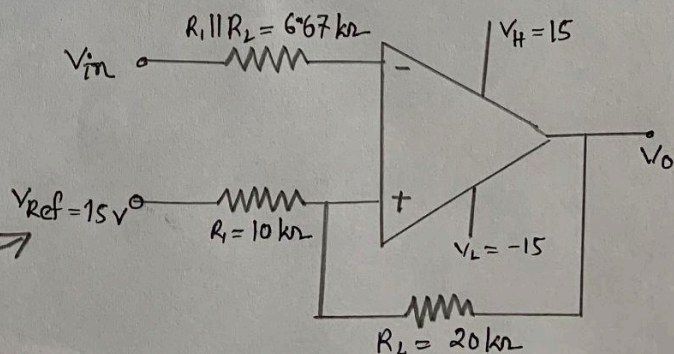
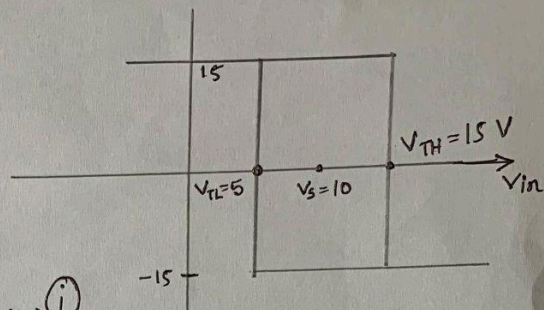
$$\text{Now, } V_S = V_{Ref} \frac{R_2}{R_1 + R_2}$$

$$\Rightarrow 10 = V_{Ref} \times \frac{20k\Omega}{10k\Omega + 20k\Omega}$$

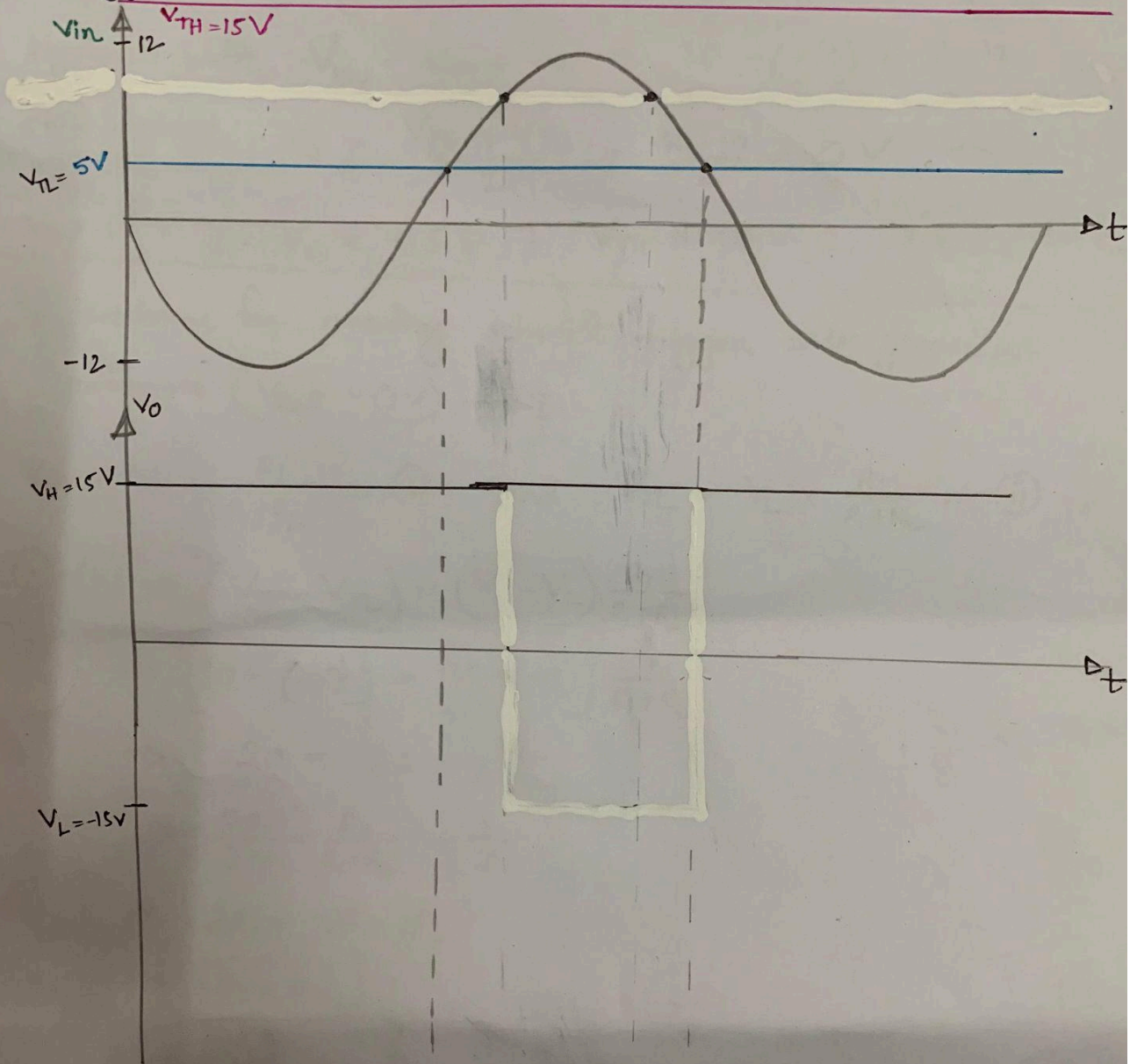
$$\Rightarrow 10 = V_{Ref} \times \frac{2}{3}$$

$$\Rightarrow V_{Ref} = \frac{10 \times 3}{2}$$

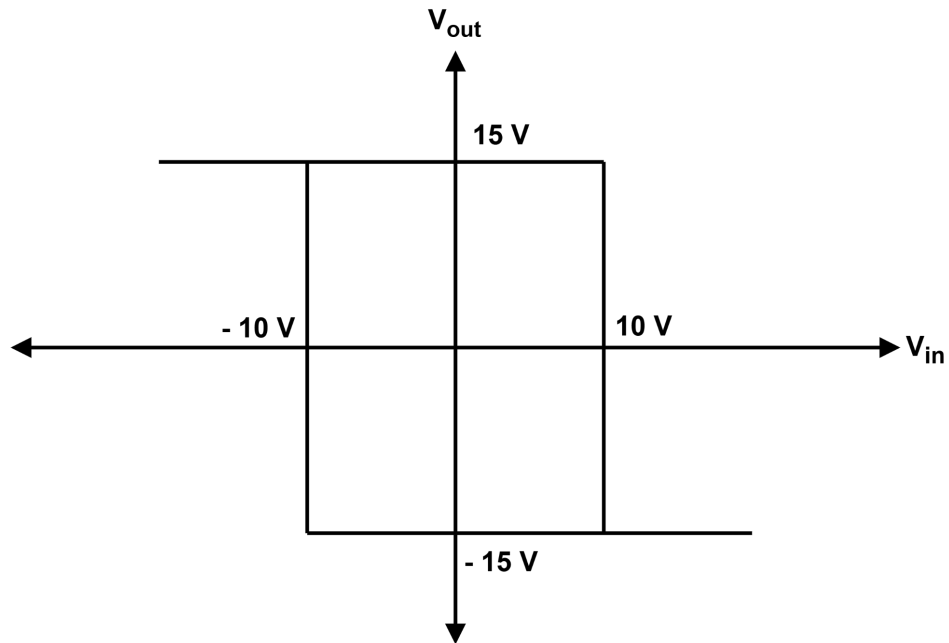
$$\boxed{V_{Ref} = 15V}$$



Applying $V_{in}(t) = -12 \sin(2\pi ft)$ into the ^{designed} Schmitt trigger and VTC derived $\rightarrow$



Set-B



(a)	CO1	Identify the Schmitt Trigger Circuit from the given VTC. What is the hysteresis width V_{HW} , shift voltage V_S , V_H , V_L , V_{TH} and V_{TL} for this specific circuit.	1+2
(b)	CO1	Design a Schmitt Trigger Circuit that will produce the same VTC.	5
(c)	CO1	Now suppose the shift voltage, $V_S = +15\text{ V}$; draw the modified VTC. Now from the modified VTC, again design the Schmitt Trigger Circuit for this case.	6
(d)	CO1	Apply an input signal $V_{in}(t) = +30\cos(2\pi ft)$ into the Schmitt Trigger and VTC derived at (c) and draw/plot the output V_{out}	6

(a) The following VTC belongs to inverting schmitt trigger with reference grounded.

$$\text{Hysteresis width, } V_{HW} = V_{TH} - V_{TL} = 10 - (-10) = 20V$$

$$\text{shift voltage, } V_S = \frac{V_{TH} + V_{TL}}{2} = \frac{10 - 10}{2} = 0V$$

$$V_H = 15V ; V_L = -15V ; V_{TH} = 10V ; V_{TL} = -10V$$

(b) we know, for inverting schmitt trigger with grounded reference ($V_{ref} = 0V$) $\rightarrow$

$$V_{TH} = V_H \times \frac{R_1}{R_1 + R_2} \dots\dots (i) \quad \text{and} \quad V_{TL} = V_L \times \frac{R_1}{R_1 + R_2} \dots\dots (ii)$$

$$(i) - (ii) \Rightarrow (V_{TH} - V_{TL}) = (V_H - V_L) \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow \{10 - (-10)\} = \{15 - (-15)\} \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow 20 = 30 \times \frac{R_1}{R_1 + R_2}$$

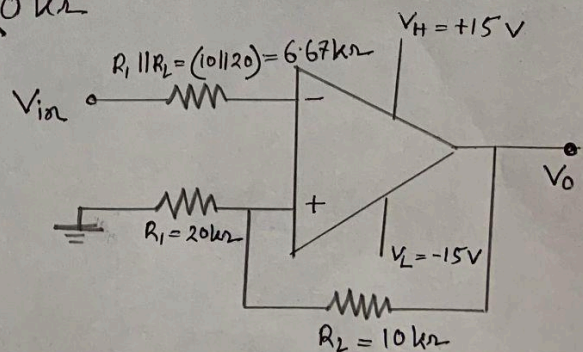
$$\Rightarrow \frac{20}{30} = \frac{R_1}{R_1 + R_2} = \frac{2}{3}$$

$$\Rightarrow 3R_1 = 2R_1 + 2R_2$$

$$\Rightarrow \boxed{R_1 = 2R_2}$$

suppose, $R_2 = 10k\Omega$; then $R_1 = 20k\Omega$

design



(c) with $V_S = +15V$, the VTC will have right shift,

Now, as there's shift voltage, so this inverting schmitt trigger has ' V_{ref} ',

we know,

$$V_{TH} = V_{ref} \times \frac{R_L}{R_1 + R_L} + V_H \times \frac{R_1}{R_1 + R_L} \dots (i)$$

$$V_{TL} = V_{ref} \times \frac{R_L}{R_1 + R_L} + V_L \times \frac{R_1}{R_1 + R_L} \dots (ii)$$

$$(i) - (ii) \Rightarrow (V_{TH} - V_{TL}) = (V_H - V_L) \frac{R_1}{R_1 + R_L}$$

$$\Rightarrow (25 - 5) = (15 + 15) \frac{R_1}{R_1 + R_L}$$

$$\Rightarrow 20 = 30 \times \frac{R_1}{R_1 + R_L}$$

$$\Rightarrow \frac{20}{30} = \frac{R_1}{R_1 + R_L} = \frac{2}{3}$$

$$\Rightarrow \boxed{R_1 = 2R_L}$$

Suppose, $R_2 = 10k\Omega$, then $R_1 = 20k\Omega$

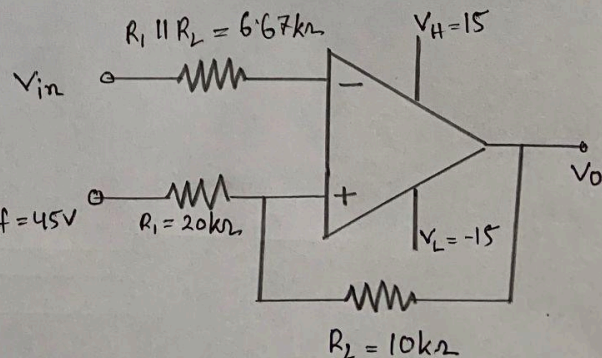
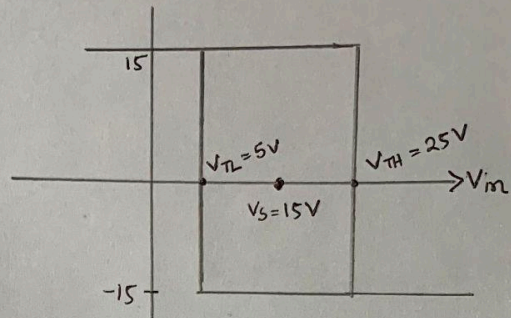
$$\text{Now, } V_S = V_{ref} \times \frac{R_L}{R_1 + R_L}$$

$$\Rightarrow 15 = V_{ref} \times \frac{10k\Omega}{20k\Omega + 10k\Omega}$$

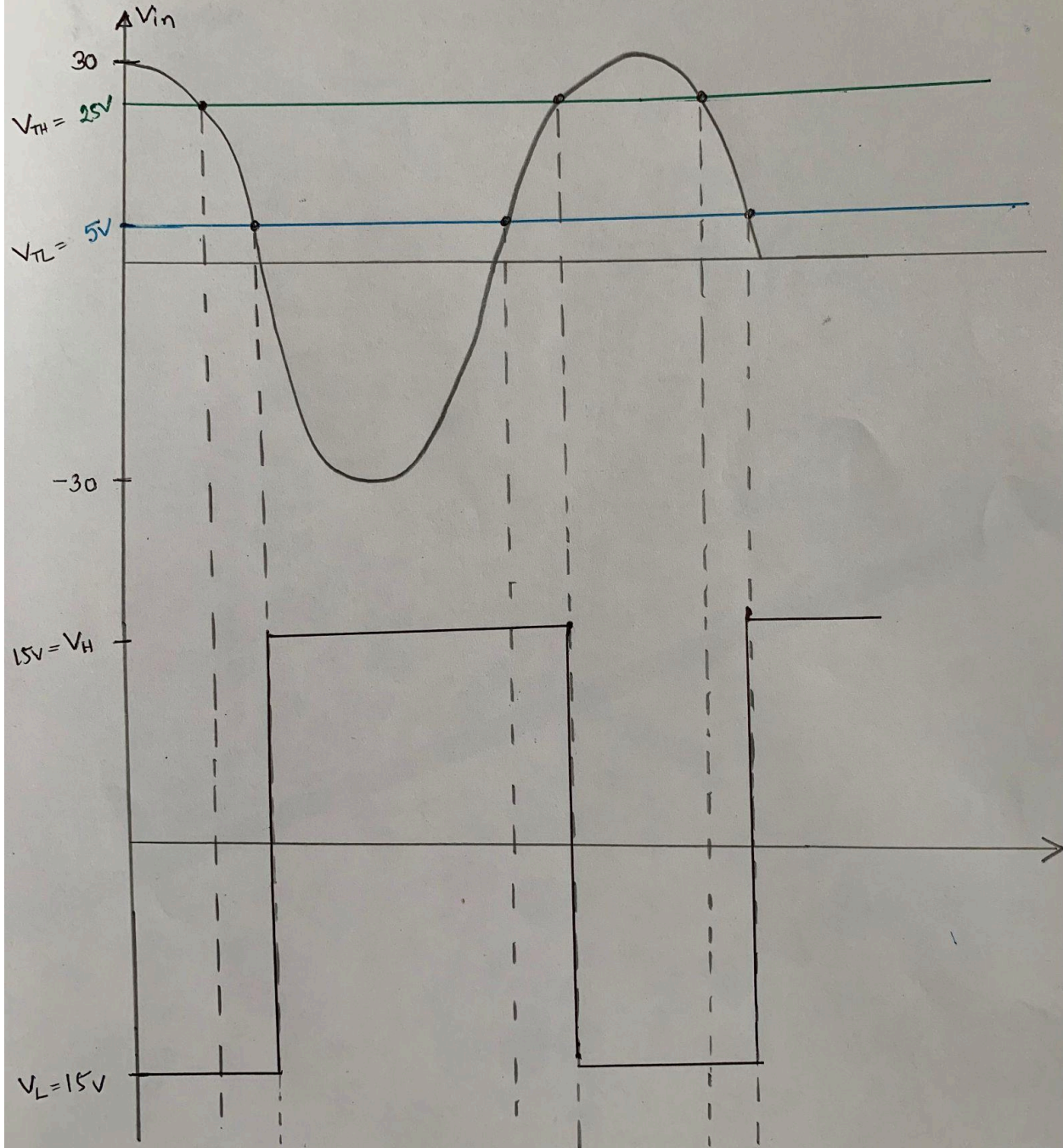
$$\Rightarrow 15 = V_{ref} \times \frac{1}{3}$$

$$\Rightarrow V_{ref} = 15 \times 3$$

$$\therefore \boxed{V_{ref} = 45V}$$

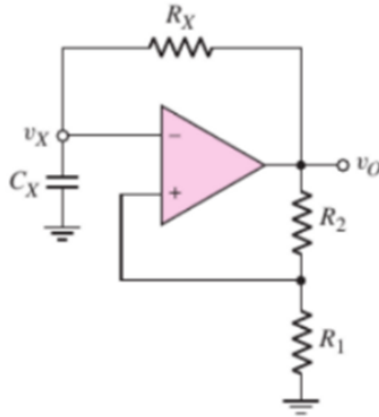


(d) Applying $V_{in}(t) = 30 \cos(2\pi ft)$ into the designed Schmitt trigger and VTC derived $\rightarrow$



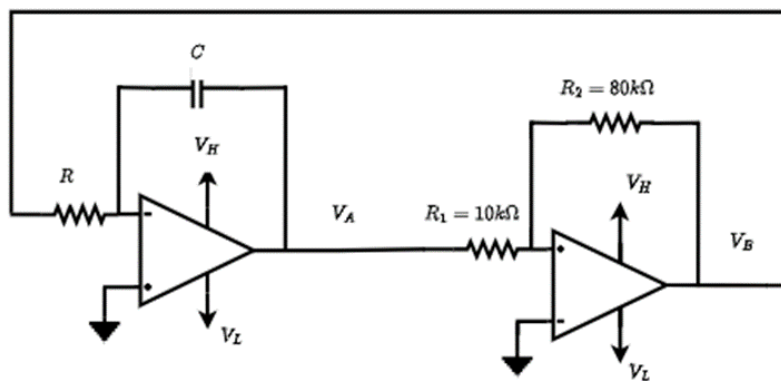
Question : Signal Generator [NUR]

Set-A



The above circuit generates a square wave with 200Hz frequency and 40% (correction 50%) duty cycle. Here, $R_1 = R_2$, $V_H = +8V$, $V_L = -8V$.

(a)	CO	If the capacitance value is $C_X = 100\text{nF}$, Find the value of R_X and Time constant.	8
(b)	CO	Draw the input-output waveform on the same graph with proper labeling on the amplitude and time axis	6



(c)	CO	For the above circuit if $V_L = -V_H$, show that the frequency will be $f = 2/RC$ (show proper calculation)	6
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Set-A (Solution)

(a) $R_1 = R_2$, $V_H = 8V$, $V_L = -8V$, $C_x = 100nF$

$f = 200Hz$, $T = \frac{1}{200} = 5ms$, $D = 0.5$

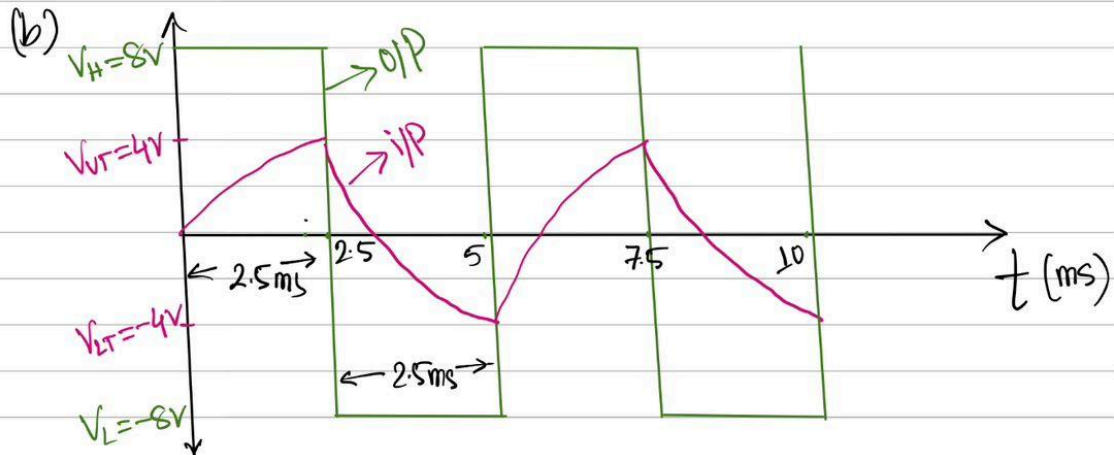
$\therefore T_1 = DT = 2.5ms$

$V_{UT} = \frac{R_1}{R_1 + R_2} V_H = 4V$, $V_{LT} = \frac{R_1}{R_1 + R_2} V_L = -4V$

$T_1 = R_x C_x \ln \left| \frac{V_H - V_{LT}}{V_H - V_{UT}} \right|$

$2.5 \times 10^{-3} = R_x \times 100 \times 10^{-9} \ln \left| \frac{8 + 4}{8 - 4} \right|$

$\boxed{R_x = 22.75K\Omega}$, Time const, $\tau = R_x C_x = 2.27ms$



(c) here $V_L = -V_H$

$V_{UT} = -\frac{R_1}{R_2} V_L = \frac{-V_L}{8} = \frac{V_H}{8}$

$V_{LT} = -\frac{R_1}{R_2} V_H = -\frac{V_H}{8}$

$T_1 = RC \left(\frac{V_{UT} - V_{LT}}{V_H} \right) = RC \left(\frac{\frac{V_H}{8} + \frac{V_H}{8}}{V_H} \right) = \frac{RC}{4}$

$T_2 = RC \left(\frac{V_{LT} - V_{UT}}{V_L} \right) = RC \left(\frac{-\frac{V_H}{8} - \frac{V_H}{8}}{-V_H} \right) = \frac{RC}{4}$

$\therefore T = T_1 + T_2 = \frac{RC}{2}$ $\therefore \boxed{\text{freq, } f = \frac{1}{T} = \frac{2}{RC}}$

Set-A (Solution)

(a) $R_1 = R_2$, $V_H = 8V$, $V_L = -8V$, $C_x = 100nF$

$f = 200Hz$, $T = \frac{1}{200} = 5ms$, $D = 0.4$

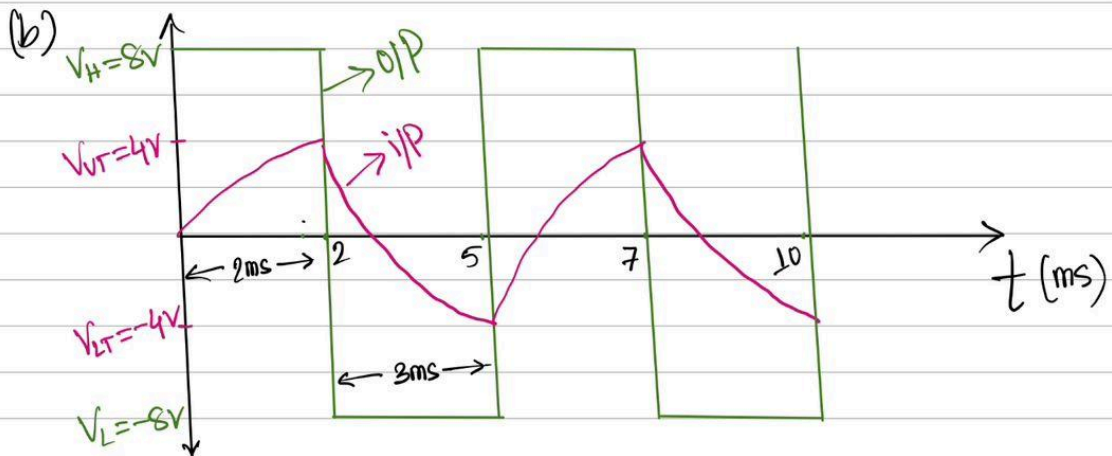
$\therefore T_1 = DT = 2ms$

$V_{UT} = \frac{R_1}{R_1 + R_2} V_H = 4V$, $V_{LT} = \frac{R_1}{R_1 + R_2} V_L = -4V$

$T_1 = R_x C_x \ln \left| \frac{V_H - V_{LT}}{V_H - V_{UT}} \right|$

$2 \times 10^{-3} = R_x \times 100 \times 10^{-9} \ln \left| \frac{8 + 4}{8 - 4} \right|$

$\boxed{R_x = 18.2 K\Omega}$, Time const, $\tau = R_x C_x = 1.82 ms$



(c) here $V_L = -V_H$

$V_{UT} = -\frac{R_1}{R_2} V_L = \frac{-V_L}{8} = \frac{V_H}{8}$

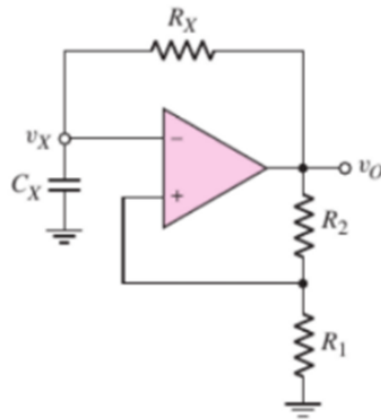
$V_{LT} = -\frac{R_1}{R_2} V_H = -\frac{V_H}{8}$

$T_1 = RC \left(\frac{V_{UT} - V_{LT}}{V_H} \right) = RC \left(\frac{\frac{V_H}{8} + \frac{V_H}{8}}{V_H} \right) = \frac{RC}{4}$

$T_2 = RC \left(\frac{V_{LT} - V_{UT}}{V_L} \right) = RC \left(\frac{-\frac{V_H}{8} - \frac{V_H}{8}}{-V_H} \right) = \frac{RC}{4}$

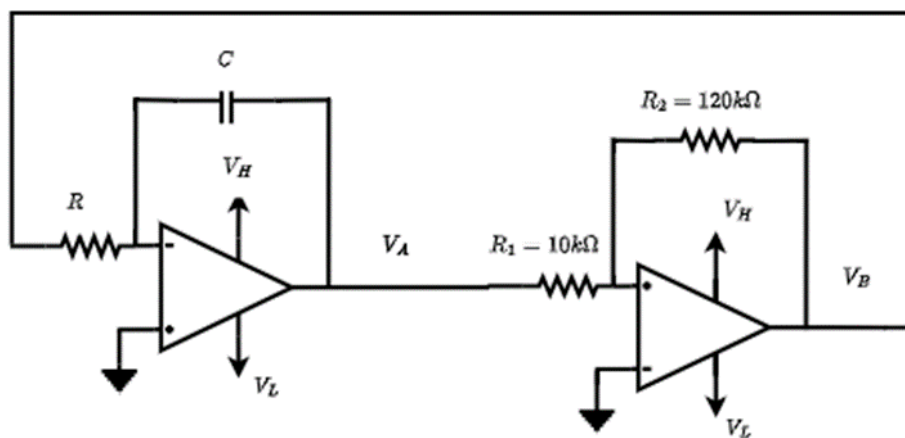
$\therefore T = T_1 + T_2 = \frac{RC}{2}$ $\therefore \boxed{\text{freq, } f = \frac{1}{T} = \frac{2}{RC}}$

Set-B



The above circuit generates a square wave with 100Hz frequency and 60% (correction 50%) duty cycle. Here, $R_1 = R_2$, $V_H = +10V$, $V_L = -10V$.

(a)	CO	If the capacitance value is $C_X = 50nF$, Find the value of R_X and Time constant.	8
(b)	CO	Draw the input-output waveform on the same graph with proper labeling on the amplitude and time axis	6



(c)	CO	For the above circuit if $V_L = -V_H$, show that the frequency will be $f = 3/RC$ (show proper calculation)	6
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Set-B (Solution)

(a) $R_1 = R_2$, $V_H = 10V$, $V_L = -10V$, $C_x = 50nF$

$f = 100Hz$, $T = \frac{1}{100} = 10ms$, $D = 0.5$

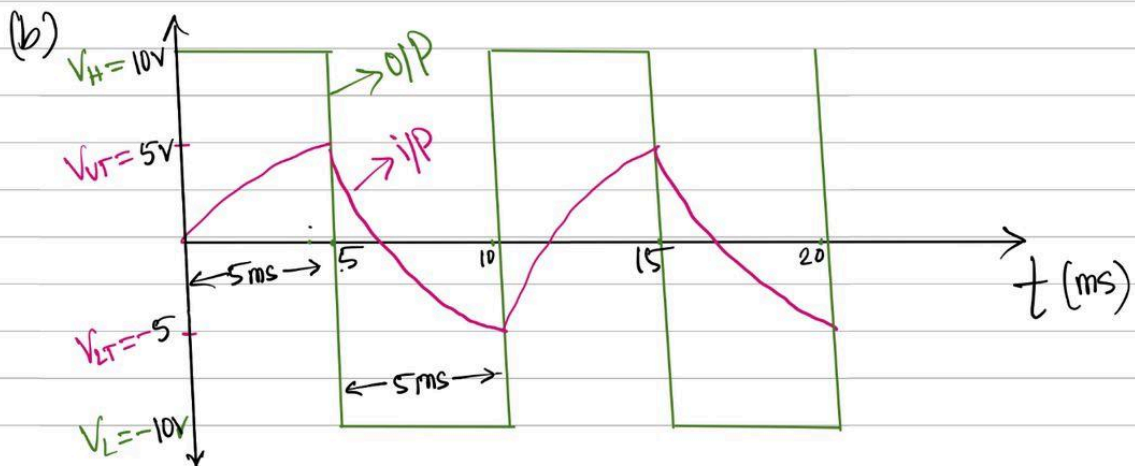
$\therefore T_1 = DT = 5ms$

$V_{UT} = \frac{R_1}{R_1 + R_2} V_H = 5V$, $V_{LT} = \frac{R_1}{R_1 + R_2} V_L = -5V$

$T_1 = R_x C_x \ln \left| \frac{V_H - V_{LT}}{V_H - V_{UT}} \right|$

$5 \times 10^{-3} = R_x \times 50 \times 10^{-9} \ln \left| \frac{10 + 5}{10 - 5} \right|$

$\boxed{R_x = 91 \text{ k}\Omega}$, Time const, $\tau = R_x C_x = 4.55ms$



(c) here $V_L = -V_H$

$V_{UT} = -\frac{R_1}{R_2} V_L = -\frac{V_L}{12} = \frac{V_H}{12}$

$V_{LT} = -\frac{R_1}{R_2} V_H = -\frac{V_H}{12}$

$T_1 = RC \left(\frac{V_{UT} - V_{LT}}{V_H} \right) = RC \left(\frac{\frac{V_H}{12} + \frac{V_H}{12}}{V_H} \right) = \frac{RC}{6}$

$T_2 = RC \left(\frac{V_{LT} - V_{UT}}{V_L} \right) = RC \left(\frac{-\frac{V_H}{12} - \frac{V_H}{12}}{-V_H} \right) = \frac{RC}{6}$

$\therefore T = T_1 + T_2 = \frac{RC}{3}$ $\therefore \boxed{\text{freq, } f = \frac{1}{T} = \frac{3}{RC}}$

Set-B (Solution)

(a) $R_1 = R_2$, $V_H = 10V$, $V_L = -10V$, $C_x = 50nF$

$f = 100Hz$, $T = \frac{1}{100} = 10ms$, $D = 0.6$

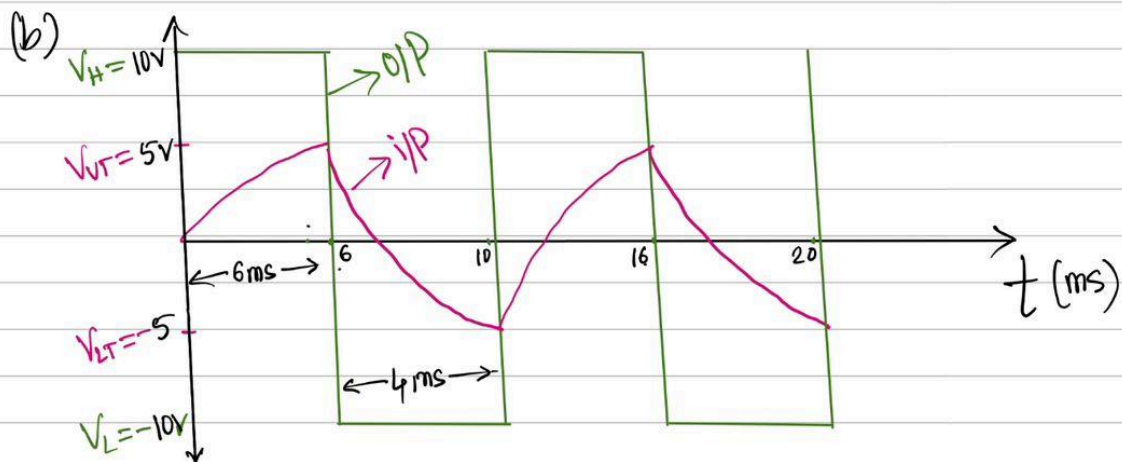
$\therefore T_1 = DT = 6ms$

$V_{UT} = \frac{R_1}{R_1 + R_2} V_H = 5V$, $V_{LT} = \frac{R_1}{R_1 + R_2} V_L = -5V$

$T_1 = R_x C_x \ln \left| \frac{V_H - V_{LT}}{V_H - V_{UT}} \right|$

$6 \times 10^{-3} = R_x \times 50 \times 10^{-9} \ln \left| \frac{10 + 5}{10 - 5} \right|$

$\boxed{R_x = 109.2k\Omega}$, Time const, $\tau = R_x C_x = 5.46ms$



(c) here $V_L = -V_H$

$V_{UT} = -\frac{R_1}{R_2} V_L = \frac{-V_L}{12} = \frac{V_H}{12}$

$V_{LT} = -\frac{R_1}{R_2} V_H = \frac{-V_H}{12}$

$T_1 = RC \left(\frac{V_{UT} - V_{LT}}{V_H} \right) = RC \left(\frac{\frac{V_H}{12} + \frac{V_H}{12}}{V_H} \right) = \frac{RC}{6}$

$T_2 = RC \left(\frac{V_{LT} - V_{UT}}{V_L} \right) = RC \left(\frac{\frac{-V_H}{12} - \frac{V_H}{12}}{-V_H} \right) = \frac{RC}{6}$

$\therefore T = T_1 + T_2 = \frac{RC}{3}$ $\therefore \boxed{\text{freq, } f = \frac{1}{T} = \frac{3}{RC}}$

Question : MOS [TSE]

Set-A

For the NMOS given in the figure,

For M1 and M3: $K_n = 0.3 \text{ mA/V}^2$ [$K_n = \frac{1}{2} \mu_n C_{ox} \frac{W}{L}$], $V_{TN} = 0.6 \text{ V}$

For M2: $K_n = 0.4 \text{ mA/V}^2$ [$K_n = \frac{1}{2} \mu_n C_{ox} \frac{W}{L}$], $V_{TN} = 0.7 \text{ V}$

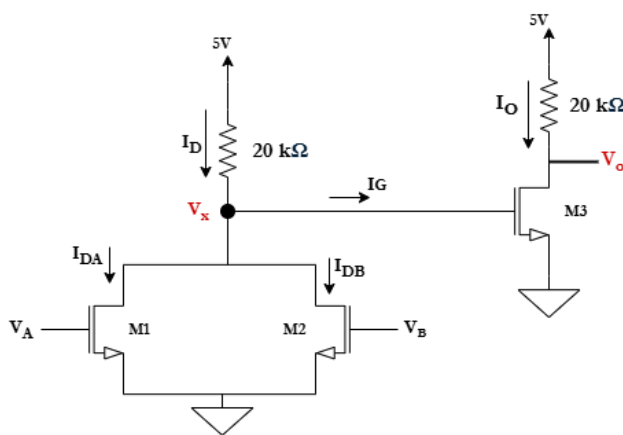


Figure 1

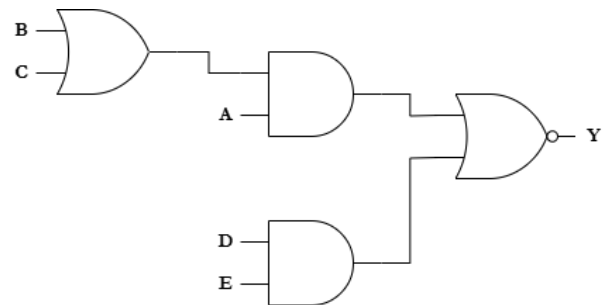


Figure 2

(a)	CO1	From the above Figure 1, determine the currents I_D , I_{DA} , I_{DB} , I_O , V_X and V_O for the cases both inputs are high and both inputs are low. (Consider 5 V as logic high and 0 V as logic low)	8
(b)	CO1	Determine the power dissipations for the cases both inputs are high and both inputs are low.	6
(c)	CO1	Find the boolean expression for output Y from Figure 2 and using that expression design an equivalent CMOS logic circuit of Figure 2.	6

Solution:

a) Both inputs are **HIGH**, so assume M1 and M2 are in triode mode.

$$I_D = I_{DA} + I_{DB}$$

$$(5 - V_X)/20 = 0.3 [2(5 - 0.6)V_X - V_X^2] + 0.4 [2(5 - 0.7)V_X - V_X^2]$$

$$> -14V_X^2 + 122.6V_X - 5 = 0$$

$$V_X = 8.72, 0.041$$

$$\text{So } V_X = 0.041 \text{ V}$$

$$I_D = 0.248 \text{ mA}$$

$$I_{DA} = 0.3 [2(5 - 0.6)(0.041) - (0.041)^2] = 0.1077 \text{ mA}$$

$$I_{DB} = 0.4 [2(5 - 0.7)(0.041) - (0.041)^2] = 0.1404 \text{ mA}$$

Since $V_X = V_{GM3} < 0.6 \text{ V}$, thus M3 is in cutoff mode

$$I_O = 0 \text{ mA}$$

$$I_G = 0 \text{ mA}$$

$$V_O = 5 \text{ V}$$

Both inputs are **LOW**, so M1 and M2 are in cutoff mode since $V_G < V_{TN}$

$$I_D = I_{DA} = I_{DB} = 0 \text{ mA}$$

$$\text{So } V_X = 5 \text{ V}$$

$$I_G = 0 \text{ mA}$$

Since $V_X = V_{GM3} = 5 \text{ V}$, thus M3 is assumed to be in triode mode

$$I_O = 0.3 [2(5 - 0.6)V_O - V_O^2]$$

$$> (5 - V_O) / 20 = 0.3 [2(5 - 0.6)V_O - V_O^2]$$

$$> -6V_O^2 + 53.8V_O - 5 = 0$$

$$V_O = 8.87 \text{ V}, 0.094 \text{ V}$$

$$\text{So, } V_O = 0.094 \text{ V}$$

$$I_O = 0.2453 \text{ mA}$$

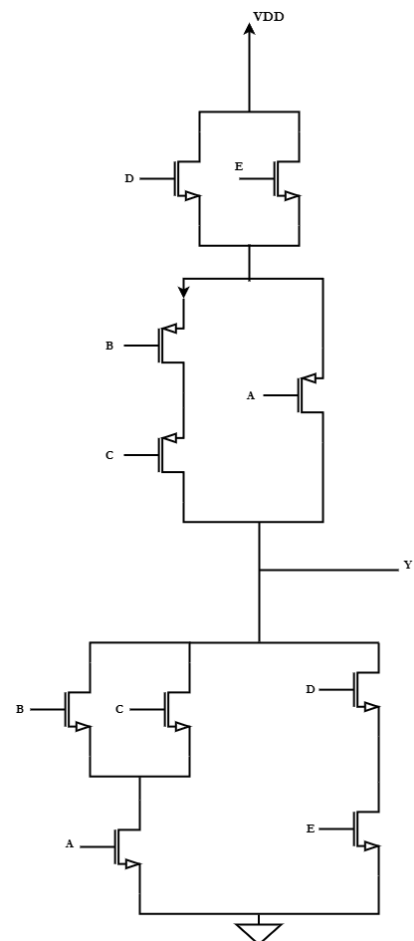
b) Power dissipation for Case (0,0)

$$P = (5 - 0) \cdot 0.2453 = 1.2265 \text{ mW}$$

Power dissipation for Case (1,1)

$$P = (5 - 0) \cdot 0.248 = 1.24 \text{ mW}$$

$$\text{c) } f = \overline{(B + C) \cdot A + D \cdot E}$$



Set-B

For the NMOS given in the figure,

For M1 and M3: $K_n = 0.4 \text{ mA/V}^2$ [$K_n = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}$], $V_{TN} = 0.8 \text{ V}$

For M2: $K_n = 0.5 \text{ mA/V}^2$ [$K_n = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}$], $V_{TN} = 0.6 \text{ V}$

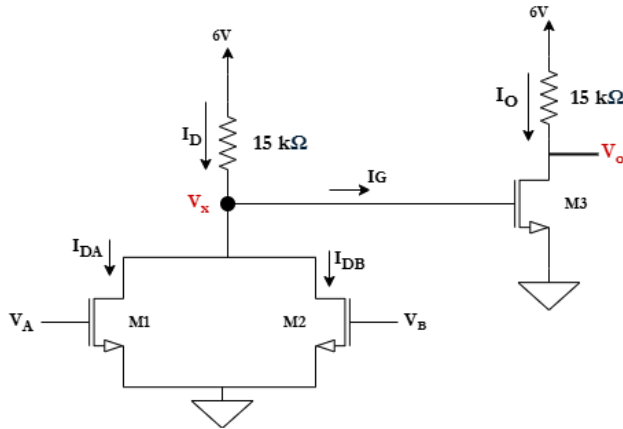


Figure 1

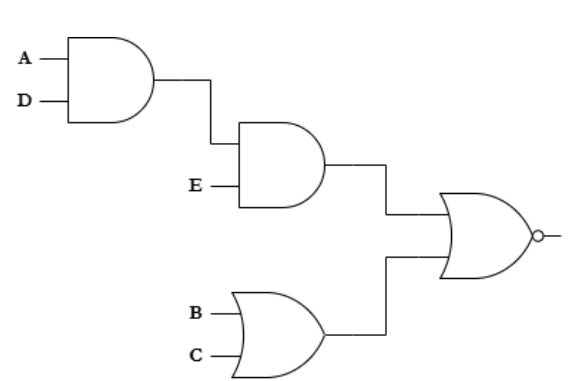


Figure 2

(a)	CO1	From the above Figure 1, determine the currents I_D , I_{DA} , I_{DB} , I_O , V_X and V_O for the cases both inputs are high and both inputs are low. (Consider 6 V as logic high and 0 V as logic low)	8
(b)	CO1	Determine the power dissipations for the cases both inputs are high and both inputs are low.	6
(c)	CO1	Find the boolean expression for output Y from Figure 2 and using that expression design an equivalent CMOS logic circuit of Figure 2.	6

Solution:

a) Both inputs are **HIGH**, so assume M1 and M2 are in triode mode.

$$I_D = I_{DA} + I_{DB}$$

$$(6 - V_X)/15 = 0.4 [2(6-0.8)V_X - V_X^2] + 0.5[2(6-0.6)V_X - V_X^2]$$

$$> -13.5V_X^2 + 144.4V_X - 6 = 0$$

$$V_X = 10.65, 0.042$$

$$\text{So } V_X = 0.042 \text{ V}$$

$$I_D = 0.3972 \text{ mA}$$

$$I_{DA} = 0.4 [2(6-0.8)(0.042) - (0.042)^2] = 0.174 \text{ mA}$$

$$I_{DB} = 0.5[2(6-0.6)(0.042) - (0.042)^2] = 0.226 \text{ mA}$$

Since $V_X = V_{GM3} < 0.6 \text{ V}$, thus M3 is in cutoff mode

$$I_O = 0 \text{ mA}$$

$$I_G = 0 \text{ mA}$$

$$V_O = 6 \text{ V}$$

Both inputs are **LOW**, so M1 and M2 are in cutoff mode since $V_G \leq V_{TN}$

$$I_D = I_{DA} = I_{DB} = 0 \text{ mA}$$

$$\text{So } V_X = 6 \text{ V}$$

$$I_G = 0 \text{ mA}$$

Since $V_X = V_{GM3} = 6 \text{ V}$, thus M3 is assumed to be in triode mode

$$I_O = 0.4 [2(6 - 0.8)V_O - V_O^2]$$

$$> (6 - V_O) / 15 = 0.4 [2(6 - 0.8)V_O - V_O^2]$$

$$> -6V_O^2 + 63.4 V_O - 6 = 0$$

$$V_O = 10.47 \text{ V}, 0.096 \text{ V}$$

$$\text{So, } V_O = 0.096 \text{ V}$$

$$I_O = 0.394 \text{ mA}$$

b) Power dissipation for Case (0,0)

$$P = (6-0) \cdot 0.394 = 2.364 \text{ mW}$$

Power dissipation for Case (1,1)

$$P = (6-0) \cdot 0.3972 = 2.3832 \text{ mW}$$

$$\text{c) } f = \frac{1}{A \cdot D \cdot E + (B + C)}$$

